

ARM PrimeCell

PL022 Synchronous Serial Port (PL022)

Errata Notice

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1 ABOUT THIS DOCUMENT

1.1 Change control

1.1.1 Change history

This refers to the revision changes to the Errata document

Issue	Date	Change
1.0	24 Oct 2001	Add Errata (Section 2.1)
1.1	14 Nov 2001	Added Errata Sections 2 and 3 Re-ordered sections i.e. Summary, Severity etc.
1.2	31 Jan 2002	Updated errata list to close REL1v1 errata

1.2 Errata List

This table summarises the errata notices detailed

Release: Release revision number affected

Section: Section in the document where the issues are described

Status: The status conditions are:

Open: Issue is current

Closed: Issue resolved in subsequent release

Severity: The severity conditions are:

A **Low** level of severity indicates that there is no direct impact to the silicon implementation. Examples are:

- No change to RTL required
- Documentation change
- Test pattern change
- Synthesis Scripts change.

A **Medium** level of severity indicates that there is a minor impact to the silicon implementation, which can be overcome by modifications detailed in the errata notes. Examples are:

- A specific sequence of events not adhered to.
- Ambiguous or unexpected operation of the device.

A **High** level of severity indicates that there is a major impact to the silicon implementation, which cannot be overcome without modifications detailed in the errata notes. Examples are:

- Non conformance to the device specification.

Affected: Deliverables impacted by the issue

Description: Issue concerning Errata

Release	Section	Status	Severity	Affected	Description
1v0	2	Closed	High	RTL	Receive Timeout Interrupt incorrectly implemented.
1v1	3	Open	Medium	Bustalk files	Incorrect Revision of Rx Timeout bustest verification files issued.
1v2	4	---	---	---	No known issues currently exist.

2 PL022 RELEASE 1V0 - ERRATA DETAILS

2.1 Receive Timeout Interrupt incorrectly implemented (pulsed observations)

2.1.1 Summary

It was discovered that the Receive Timeout period was being timed in units of the SSPCLK periods when it should be in terms of bit periods.

2.1.2 Severity

High

There is no workaround to this problem. The Verilog and VHDL RTL must be corrected in order for this function to be used in the proper manner.

2.1.3 Symptoms

A symptom of this error is a pulsing of the Receive Timeout interrupt signals occurring after the end of each received word when SSPCLK is a relatively high frequency in comparison with the data bit rate.

2.1.4 Affected items

A minor modification to the state machine code within the SspMTxRxCntl block has been made to correct the operation of the Receive Time Out interrupt generation and new tests written to verify the changes.

RTL source files affected: Verilog SspMTxRxCntl.v and Ssp.v, VHDL SspMTxRxCntl.vhd and Ssp.vhd

Additional "receive time out" test patterns have been added to those already present in REL1v0 to test that the fix has been implemented correctly.

Verification bustest files created to further test the Receive Timeout interrupt function:

Ssp.c	Ssp.h
Ssp_Mw_RTIS_Test.c	Ssp_Mw_RTIS_Test_S.c
Ssp_Mw_rtis_Interrupt_Test.c	Ssp_Mw_rtis_Interrupt_Test_S.c
Ssp_Spi00_RTIS_Test.c	Ssp_Spi00_RTIS_Test_S.c
Ssp_Spi00_rtis_Interrupt_Test.c	Ssp_Spi00_rtis_Interrupt_Test_S.c
Ssp_Spi01_RTIS_Test.c	Ssp_Spi01_RTIS_Test_S.c
Ssp_Spi01_rtis_Interrupt_Test.c	Ssp_Spi01_rtis_Interrupt_Test_S.c
Ssp_Spi10_RTIS_Test.c	Ssp_Spi10_RTIS_Test_S.c
Ssp_Spi10_rtis_Interrupt_Test.c	Ssp_Spi10_rtis_Interrupt_Test_S.c
Ssp_Spi11_RTIS_Test.c	Ssp_Spi11_RTIS_Test_S.c
Ssp_Spi11_rtis_Interrupt_Test.c	Ssp_Spi11_rtis_Interrupt_Test_S.c
Ssp_Ti_RTIS_Test.c	Ssp_Ti_RTIS_Test_S.c
Ssp_Ti_rtis_Interrupt_Test.c	Ssp_Ti_rtis_Interrupt_Test_S.c

The tests should be compiled using the “makefile” released with SSP PL022 REL1v0.

2.1.5 Short term work around

There is no short term software workaround.

2.1.6 Proposed Corrective action

Both the VHDL and Verilog SspMTxRxCntl and Ssp RTL source files must be corrected as described in the following sections.

A maintenance release **SSP PL022 REL1v1** incorporating these changes is available.

For VHDL, the changes required to fix the bug are:

File SspMTxRxCntl.vhd

From : Line 69: RTIMSync : in std_logic; -- Rx TimeOut Mask
Line 70: --

To : Line 69: RTIMSync : in std_logic; -- Rx TimeOut Mask
Line 70: RNESync : in std_logic; -- Rx FIFO Not Empty

From : Line 385 RTIMSync, BitPeriodCnt, BitCnt, TxShft, RxShft,
To : Line 385 RTIMSync, RNESync, BitPeriodCnt, BitCnt, TxShft, RxShft,

From : Line 383: NextIncRxTimeOut <= ilncRxTimeOut;
To : Line 383: NextIncRxTimeOut <= '0';

From : Lines 403, 461, 763 and 1045 : if ((SSPCLKDIV and BitPeriodCmp) = '1') then
To : Lines 403, 461, 763 and 1045 : if ((SSPCLKDIV and BitPeriodCmp and RNESync) = '1') then

File Ssp.vhd

From : Line 352 : SSPCLKDIV or RTIMSync or BitPeriodCnt or BitCnt or
To : Line 352 : SSPCLKDIV or RTIMSync or RNESync or BitPeriodCnt or BitCnt or

From : Line 505: RTIMSync : in std_logic;
Line 506:

To : Line 505: RTIMSync : in std_logic;
Line 506: RNESync : in std_logic;

From : Line 1476: RTIMSync => RTIMSync,
Line 1477:

To : Line 1476: RTIMSync => RTIMSync,
Line 1477: RNESync => RNESync,

For Verilog, the changes required to fix the bug are:

File SspMTxRxCntl.v

From : Line 51: RTIMSync,
Line 52:

To : Line 51: RTIMSync,
Line 52: RNESync,

From : Line 87: input RTIMSync; // Rx TimeOut Mask

To : Line 87: input RTIMSync; // Rx TimeOut Mask
Line 88: input RNESync; // RxFIFO Not Empty

From : Line 370: NextIncRxTimeOut = IncRxTimeOut;

To : Line 370: NextIncRxTimeOut = 1'b0;

From : Lines 390, 457, 806 and 1156: if (SSPCLKDIV & BitPeriodCmp)

To : Lines 390, 457, 806 and 1156: if (SSPCLKDIV & BitPeriodCmp & RNESync)

File Ssp.v

From:

Line 950: .RTIMSync (RTIMSync),
Line 951

To:

Line 950: .RTIMSync (RTIMSync),
Line 951: .RNESync (RNESync),

2.2 Technical Reference Manual Documentation Errata

2.2.1 Summary

ARM DDI 0194 (Rev A,B,C,D) SSP PL022 Technical Reference Manual

1. Page 2-18, 2.3.13 Motorola SPI Format with SPO=1, SPH=1.
“SPO=0” is incorrect in the first sentence; this should be “SPO=1”.
2. Page 2-18, Figure 2-9 Motorola SPI Format with SPO=1, SPH=1.
The SSPCLKOUT/SSPCLKIN waveform is incorrect. The clock sequence should terminate after 8 clock and return to a high state.

2.2.2 Severity

Low

These errors are issued as documentation errata and do not affect the operation or verification of the device.

2.2.3 Symptoms

1. Typing mistake.
2. The waveform will not match simulation or real silicon waveforms.

2.2.4 Affected items

ARM DDI 0194 (Rev A,B,C,D) SSP PL022 Technical Reference Manual

2.2.5 Short term work around

Published as documentation errata.

2.2.6 Proposed Corrective action

To be corrected in next release of the ARM DDI 0194 SSP PL022 Technical Reference Manual – no date available.

3 PL022 RELEASE 1V1 - ERRATA DETAILS

3.1 Incorrect revision of Rx Timeout bustest verification files issued

3.1.1 Summary

The verification tests were originally written to validate the Receive Timeout interrupt feature of the PrimeCell SSP PL022 REL1v0. It was subsequently reported that the Receive Timeout was generated after 32 periods of the input SSPCLK and not that of the bit period.

The Verilog and VHDL RTL code was modified to correct this and released as REL1v1. Improved test patterns were written to further test this feature.

However, the new tests were mistakenly not checked into the release database, which did not allow the SSP PL022 REL1v1 to be independently re-verified at a customer site.

3.1.2 Severity

Medium

The Verilog and VHDL rtl is correct, but the shipping of the incorrect test files inhibits the customer from verifying the deliverables using the standard 'Out of box' procedure within the "acceptance period".

3.1.3 Symptoms

The shipped SSP PL022 REL1v1 ARM RTL/Netlist datalogs contain no reported simulation error messages.

However, if the customer reruns all the verification test patterns, then it will be evident that the simulations fail with errors messages similar to those shown below being reported.

This is symptomatic that the incorrect revision of the Rx Timeout bustest verification test source files have been mistakenly shipped to the customer.

Examples of such error messages are:

```
# Time: 565816 ns TEST SPI11 RTIS
# 642540 ns: POE: Maximum number of Polls for value 00000410 reached. TAG: poll_rt_deasserted
# 642570 ns: PSRE: Error on data read. Expected: 00000410 Actual: 00001464 Mask: 00001fff TAG: rd_rt_deasserted
# 858430 ns: POE: Maximum number of Polls for value 00000444 reached. TAG:poll_2nd_tx_complete
# 858460 ns: PSRE: Error on data read. Expected: 00000444 Actual: 00001464 Mask: 00001fff TAG: rd_2nd_tx_complete
# 860160 ns: PSRE: Error on data read. Expected: 00000000 Actual: 0000000e Mask: 00000002 TAG: rd_sspris
# 860180 ns: PSRE: Error on data read. Expected: 00000000 Actual: 00000002 Mask: 00000002 TAG: rd_sspmis
# Time: 860646 ns SPI11 RTIS test over
# Time: 860786 ns RTIS Interrupt Test End
# Time: 860786 ns TI RTIS Interrupt tests
# 938420 ns: POE: Maximum number of Polls for value 00000410 reached TAG: poll_rt_deasserted
```

```
# 938450 ns: PSRE: Error on data read. Expected: 00000410 Actual: 00001464 Mask: 00001fff TAG: rd_rt_deasserted
# 1154310 ns: POE: Maximum number of Polls for value 00000444 reached. TAG: poll_2nd_tx_complete
# 1154340 ns: PSRE: Error on data read. Expected: 00000444 Actual: 00001464 Mask: 00001fff TAG: rd_2nd_tx_complete
# 1156040 ns: PSRE: Error on data read. Expected: 00000000 Actual: 0000000e Mask: 00000002 TAG: rd_sspris
# 1156060 ns: PSRE: Error on data read. Expected: 00000000 Actual: 00000002 Mask: 00000002 TAG: rd_sspmismis
# Time: 1156526 ns TI RTIS test over
# Time: 1156666 ns TI RTIS Interrupt Test End
# Time: 1156666 ns MW RTIS Interrupt tests
# Time: 1156666 ns MW RTIS INTERRUPT TEST
# Time: 1156666 ns Baud 1, Wordlength 16, RXW_4
# Time: 1157556 ns TEST MW RTIS
# 1163110 ns: PSRE: Error on data read. Expected: 00000410 Actual: 00000450 Mask: 00001fff TAG: rd_rt_deasserted
# 1166740 ns: PSRE: Error on data read. Expected: 00000000 Actual: 0000000e Mask: 00000002 TAG: rd_sspris
# 1166760 ns: PSRE: Error on data read. Expected: 00000000 Actual: 00000002 Mask: 00000002 TAG: rd_sspmismis
```

3.1.4 Affected items

The following Bustalk test source files should be copied into the existing ssp_pl022/verification/bustest directory such that they overwrite/replace those issued with the SSP PL022 REL1v2 release:

Ssp.c	Ssp.h
Ssp_Mw_RTIS_Test.c	Ssp_Mw_RTIS_Test_S.c
Ssp_Mw_rtis_Interrupt_Test.c	Ssp_Mw_rtis_Interrupt_Test_S.c
Ssp_Spi00_RTIS_Test.c	Ssp_Spi00_RTIS_Test_S.c
Ssp_Spi00_rtis_Interrupt_Test.c	Ssp_Spi00_rtis_Interrupt_Test_S.c
Ssp_Spi01_RTIS_Test.c	Ssp_Spi01_RTIS_Test_S.c
Ssp_Spi01_rtis_Interrupt_Test.c	Ssp_Spi01_rtis_Interrupt_Test_S.c
Ssp_Spi10_RTIS_Test.c	Ssp_Spi10_RTIS_Test_S.c
Ssp_Spi10_rtis_Interrupt_Test.c	Ssp_Spi10_rtis_Interrupt_Test_S.c
Ssp_Spi11_RTIS_Test.c	Ssp_Spi11_RTIS_Test_S.c
Ssp_Spi11_rtis_Interrupt_Test.c	Ssp_Spi11_rtis_Interrupt_Test_S.c
Ssp_Ti_RTIS_Test.c	Ssp_Ti_RTIS_Test_S.c
Ssp_Ti_rtis_Interrupt_Test.c	Ssp_Ti_rtis_Interrupt_Test_S.c

The tests should then be re-compiled using the “makefile” released with SSP PL022 REL1v1.

All new logs that are generated as a result of the subsequent rtl and netlist simulations should then replace those shipped with the SSP REL1v1 release.

3.1.5 Short term work around

Updated files available from support-primecell@arm .com

3.1.6 Proposed Corrective action

Files have been entered into the release database. A new release, PL022 REL1v2 is to be made available

In conjunction, the REL1v5 "global" files will accompany the SSP REL1v2 release and the this combination should be used together for subsequent out-of-box verification purposes.

4 PL022 RELEASE 1V2 - ERRATA DETAILS

4.1 No known issues

The PL022 REL1v2 release contains the bustest files that were mistakenly not checked for the REL1v1 release. The rtl and netlist simulations using the full set of correct test pattern files have been performed and the respective log files are included in the REL1v2 release.